



Smart Recruitment Assistant

TEAM MEMBERS

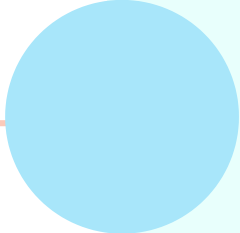
Basmalah Ayman

Rawan Ayman

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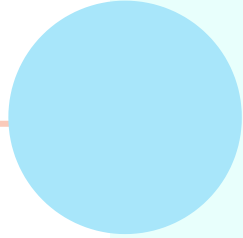
Maryam Ahmed

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Problem Statement & Objectives

- Problem: Difficulty in predicting employees or trainees who are considering leaving their jobs or looking for new opportunities.
- Objective: Build a Machine Learning model to predict whether a person will look for a new job or not.
- Value: Assist the HR department in retaining talent and reducing recruitment and training costs.

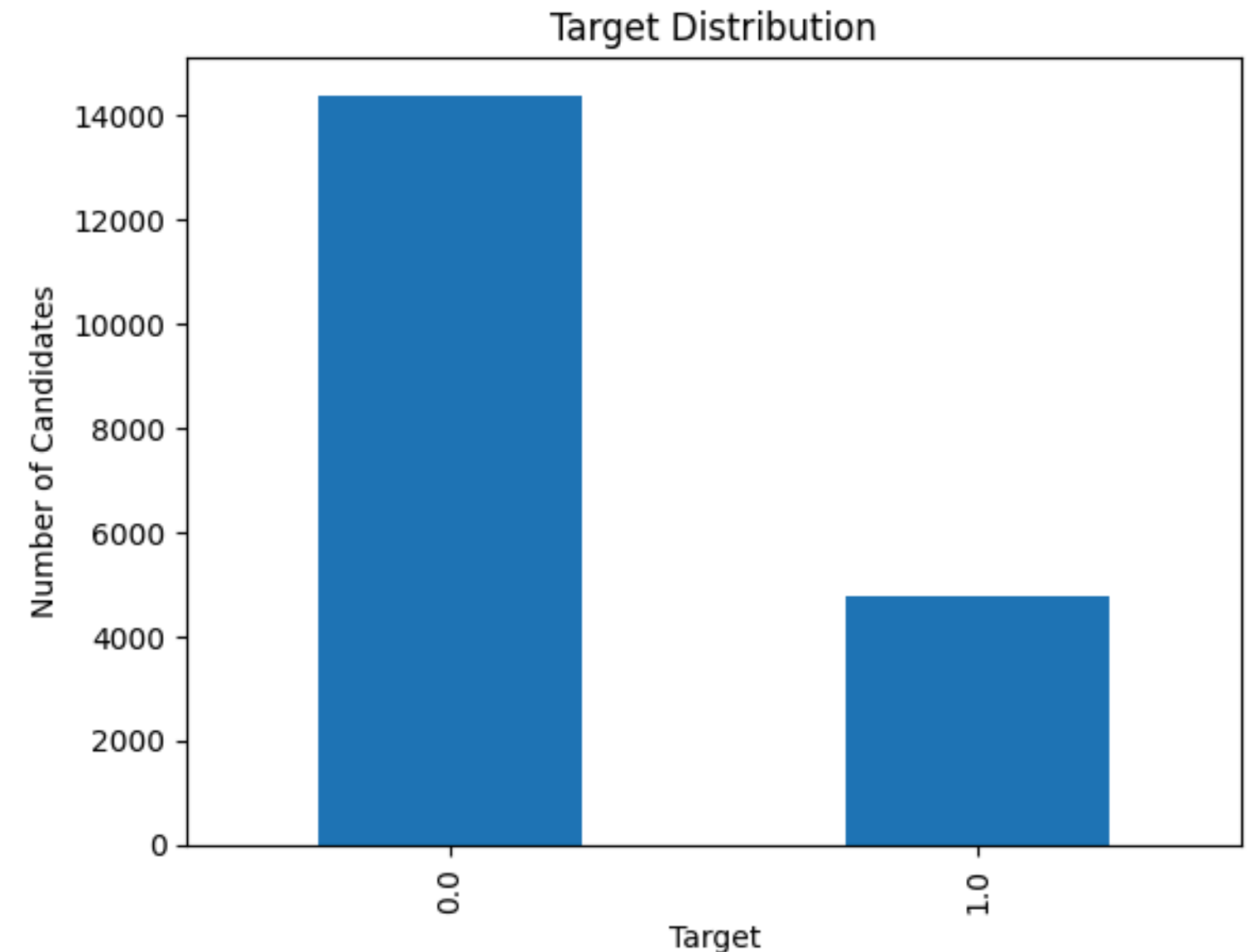


Dataset Summary

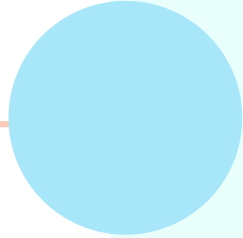
- Shape : (19158, 14)
- Duplicate Rows: No duplicate rows found in the dataset.
- Missing values exist in :
gender , enrolled_university university ,
educational_level , major_discipline ,
experience , company_size ,
company_type , last_new_job

Target Distribution Analysis

- 75% of individuals are not looking for a job change (14,381 cases).
- 25% of individuals are looking for a job change (4,777 cases).

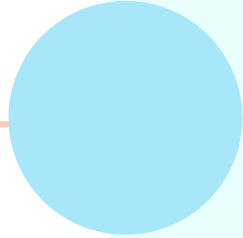


Key Observation: The dataset exhibits class imbalance, which was taken into consideration during model evaluation.



Data Preprocessing

- Missing Values handling : Used mean to fill missing values
- Categorical Encoding: Converted categorical variables into numerical format using OneHotEncoder.
- Feature Scaling: Standardized numerical features using StandardScaler.
- Pipeline Integration: Combined all steps into a Scikit-learn Pipeline to prevent data leakage.
- Outliers handled using capping method .
- Data is balanced using Random Over-Sampling (ROS).



Machine Learning Models

Data was split into Training and Testing sets.

Data were Trained and compared using multiple classification models:

- Logistic Regression
- SVM
- Decision Tree
- Random Forest
- K-Nearest Neighbors (KNN)

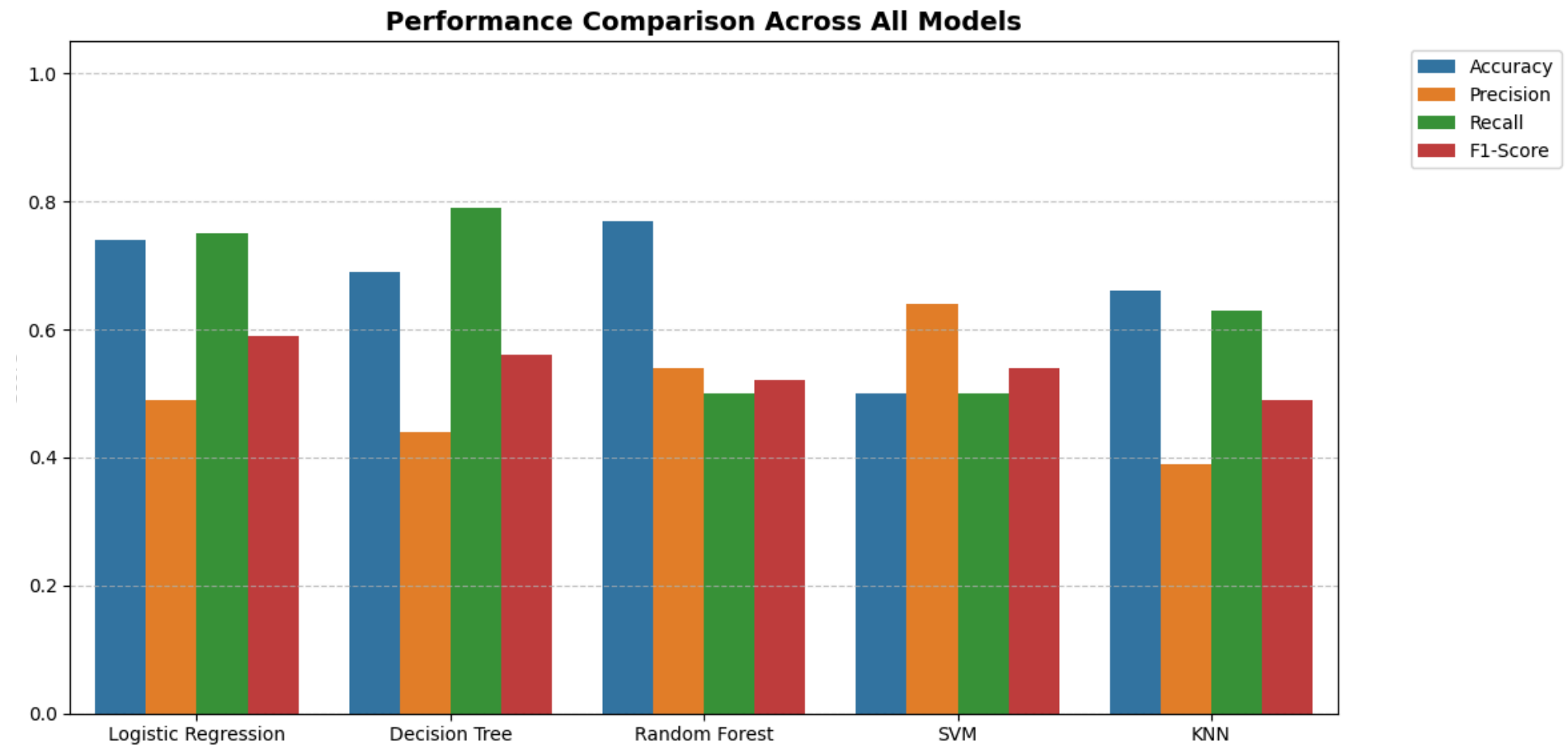
Results & Model Comparison

A summary table comparing model performance metrics: Accuracy, Precision, Recall, and F1-Score.

Why Logistic Regression Model?

Because it is the one who has Achieved the best balance in predicting F1 score
As in my data what affects the most is False info (FP , FN)

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$



BestModel:LogisticRegression

THANK
YOU